

## Article

# 4D Seismic Monitoring with Diffraction-Angle-Filtering for Carbon Capture and Storage (CCS)

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**Abstract:** Carbon capture and storage (CCS) is a key technology that directly removes industry driven CO<sub>2</sub> to achieve carbon neutrality. In the process of CCS, it is necessary to monitor whether injected CO<sub>2</sub> is properly stored and not leaking. The behavior of CO<sub>2</sub> can be investigated using a 4D seismic survey that compares seismic data before and after injection. We proposed a two-step monitoring with diffraction-angle filtering (DAF) to effectively locate the CO<sub>2</sub> plume. Because DAF allows us to control wavenumber components, the gradient of full-waveform inversion (FWI), which is the first step, is composed of low-wavenumber components, and reverse time migration (RTM) for seismic imaging is carried out with high-wavelength components. To verify our method, we implemented FWI and RTM with and without DAF using the velocity model in the Volve oil field in the North Sea. Numerical examples show that the CO<sub>2</sub> plume is properly detected from the difference between baseline and post-injected survey and the extension of the reflective boundary is improved compared to the results of the conventional method. With our proposed method, local minima problem is mitigated in FWI, and the boundaries between layers can be clearly distinguished in RTM.

**Keywords:** seismic monitoring; carbon capture and storage; wave scattering and diffraction

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## 1. Introduction

Carbon capture and storage (CCS) is a series of technologies used to capture a large amount of CO<sub>2</sub> and store it in a geological storage site. CO<sub>2</sub> is a representative greenhouse gas that increases atmospheric temperature by absorbing infrared radiant heat and contributes the most to global warming among greenhouse gases [1]. The problem is that CO<sub>2</sub> emissions have risen dramatically in this century. The net anthropogenic cumulative CO<sub>2</sub> emissions from 2010 to 2019 were about 410 Gt CO<sub>2</sub>, accounting for 17% of those from 1850 to 2019. Although other factors are involved in changes in temperature such as solar radiation, as a result, the surface temperature in 2021 was 1.04 °C warmer than the pre-industrial period (1880–1900), and it is expected to increase to 3.2 °C in 2100 [1]. Because the heated Earth causes a decrease in snow cover and sea ice, heavy rainfall, extremely high temperature and changes in the habitats of living things, the global warming target was set to rise by 1.5 °C compared to the pre-industrial period at COP 21 [2]. According to COP 26, the parties should reduce CO<sub>2</sub> emissions by 45% compared 2010 by 2030 and carbon neutrality, which refers to a net-zero state of CO<sub>2</sub> emission, must be achieved by 2050 to meet the target temperature. For carbon neutrality, CCS plays an important role in directly reducing industry driven CO<sub>2</sub>. Increasing the use of renewable energy is essential to mitigate global warming in the long run, but it is difficult to achieve carbon neutrality within 2050 without CCS in the current industrial structures. It is expected that CCS alone could contribute 13% of the CO<sub>2</sub> mitigation needed worldwide by 2050 [3]. As the number of projects has increased due to the demand of CCS, the overall capacity of

CCS has increased by about four times in 2022 compared to 2017, and will continue to rise [4]. In 1972, the first commercial scale project, the Scurry Area Canyon Reef Operating Committee (SACROC), was implemented, and 175 million tons of CO<sub>2</sub> was injected for enhanced oil recovery (EOR) purposes [5]. The first large-scale project for reducing anthropogenic CO<sub>2</sub> emissions was launched in Norway in 1996, which was the Sleipner CCS project. They captured CO<sub>2</sub> from the natural gas produced in Sleipner west field and injected the separated CO<sub>2</sub> into a deep saline reservoir of 800 to 1000 m. Furthermore, 3D seismic monitoring has been performed for 20 consecutive years, and there is no evidence of leakage of the injected CO<sub>2</sub> into other horizons [6,7]. Since the success of the Sleipner project, several projects aimed at underground storage have been implemented, including the Weyburn-Midale [8] and the Quest project in Canada [9], and the Illinois project in the United States [10,11].

Geological storage site for CCS can be divided into onshore and offshore. The storage in the marine environment is advantageous in countries with unaffordable territory or in depleted oil and gas reservoirs and saline aquifers projects [12]. In addition, a positive feature of offshore projects is that they can increase public acceptance because most of them are located far from residential areas [13]. However, the cost of facilities for transportation, injection, and storage are more expensive than the cost of onshore projects, and there are some technical issues that require marine engineering to address. The seawater temperature, which can drop to around 0 °C in winter, affects the injection process. Even though the temperature is sufficiently increased at the inlet of the injection well, it may drops to the seawater temperature as the injection pipes pass through sea water, which causes the conversion to CO<sub>2</sub> hydrate [14]. Because the density of supercritical gas changes depending on the temperature, it is important to obtain accurate temperature data for each section in injection well to understand the behavior of CO<sub>2</sub>. Moreover, offshore monitoring for CCS is also an important consideration. The reason for monitoring is to investigate whether CO<sub>2</sub> is injected in the correct location or is not leaking. Escaped CO<sub>2</sub> can acidify sea water and has negative effects on the surrounding marine ecosystem. Due to the complexity of the marine environment, monitoring requires remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs) with sensors for direct detection, and seismic exploration in indirect ways [15].

Seismic exploration is a geophysical method that detects reflected or refracted seismic waves from subsurface media using artificial sources. It has been mainly used to detect hydrocarbons because of the different physical properties of gas and fluid compared to those of the surrounding rocks and the resulting high-resolution images. For the same reason, seismic exploration can contribute to site selection and monitoring for CCS. A storage site composes reservoirs such as saline aquifers and caprocks, which is similar to the structure of petroleum trap. Drilling provides the exact values of physical properties in several spots, but a wide range of structures can be constructed by seismic exploration. High-resolution images and physical properties obtained by seismic data processing provide information on storage capacity, location of the reservoir and caprock, and the distribution of adjacent fractures.

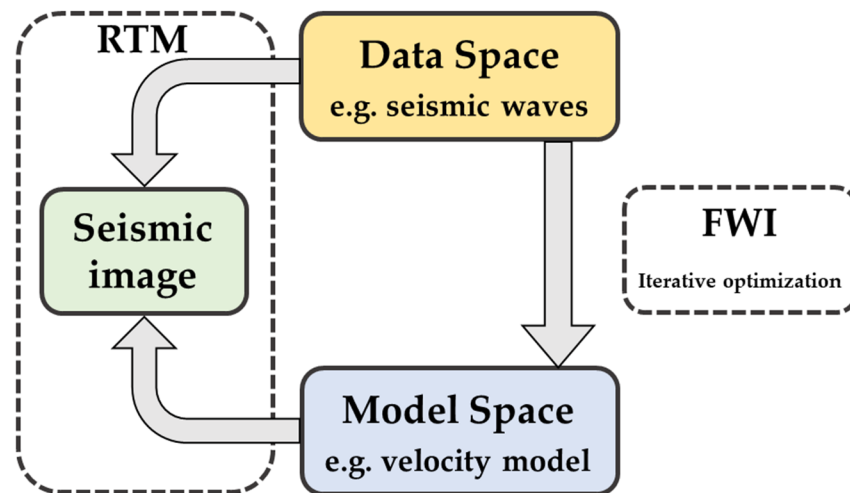
Seismic monitoring for CCS compares seismic data before and after CO<sub>2</sub> injection, and is called 4D monitoring. CO<sub>2</sub> has different physical properties from the surrounding rocks, resulting in a difference in seismic impedance, and we can investigate the behavior of stored CO<sub>2</sub> and leaking from seismic data. An effective way to compare the state between pre- and post-injection is to analyze the seismic images. Seismic imaging constructs the boundary of layers having different physical properties with seismic data, and the reverse time migration (RTM) is a seismic imaging method based on the two-way wave equation, which has high computational cost but represents the complex structures well [16]. Conventional imaging uses the velocity obtained by tomography and locates the reflectors with short-wavelength components. Instead of tomography, full-waveform inversion (FWI) [17] has been used to obtain more accurate and high-resolution velocity models for decades. However, FWI is vulnerable to the local minima problem when long-

wavelength components are not properly established [18]. In this paper, we propose a two-step monitoring approach with diffraction-angle-filtering (DAF) [19] to improve the quality of seismic images and monitoring results. Because DAF allows us to control the scattering angle, we can use the long-wavelength component for FWI to mitigate local minima and the short wavelengths for RTM. In the following section, we review the basic theory of conventional seismic FWI and RTM and introduce a two-step monitoring with DAF. Then, we verify our method through synthetic tests in the Volve oil field.

## 2. Methods

### 2.1. Seismic Imaging

As mentioned above, 3D/4D seismic imaging is an essential tool to find subsurface CO<sub>2</sub> storage and to monitor CO<sub>2</sub> leakage. Seismic imaging based on RTM requires numerical modeling to obtain forward and backward wavefields. The velocity model obtained by FWI is an important factor in accurate modelling. Figure 1 shows the process of seismic imaging between the data and model spaces using FWI and RTM. We review the conventional FWI for velocity reconstruction, RTM for imaging and 4D monitoring for CCS in this section.



**Figure 1.** Schematic process of seismic imaging between data and model space using FWI and RTM.

#### 2.1.1. Acoustic FWI

Seismic FWI estimates physical properties by minimizing the objective function that consist of the residual between observed and modeled data as follows:

$$E(\mathbf{m}) = \sum_{\omega} \sum_s \frac{1}{2} \left[ \tilde{\mathbf{p}}(\omega, s, \mathbf{m}) - \tilde{\mathbf{d}}(\omega, s) \right]^2, \quad (1)$$

where  $\tilde{\mathbf{p}}$  and  $\tilde{\mathbf{d}}$  are calculated and observed pressure wavefields in the frequency domain, respectively.  $\omega$  and  $s$  indicate angular frequency and source, respectively, and  $\mathbf{m}$  are the model parameters including P-wave velocity ( $v_p$ ). The gradient direction of P-wave velocity can be calculated by taking a partial derivative of Equation (1) with respect to P-wave velocity as follows:

$$\frac{\partial E(\mathbf{m})}{\partial m_i} = \text{Re} \left[ \sum_{\omega} \sum_s \left\{ \left( \frac{\partial \tilde{\mathbf{p}}(\omega, s, \mathbf{m})}{\partial m_i} \right)^T \cdot (\tilde{\mathbf{p}}(\omega, s, \mathbf{m}) - \tilde{\mathbf{d}}(\omega, s)) \right\} \right], \quad (2)$$

where  $m_i$  indicate the model parameter at the  $i$ th nodal point. To reduce the large computation of Equation (2), we adopt the adjoint-state method using virtual sources [20]. Using the adjoint-state method, the gradient direction can be rewritten as:

$$\frac{\partial E(\mathbf{m})}{\partial m_i} = \text{Re} \left[ \sum_{\omega} \sum_s \left\{ \mathbf{f}_i^v(\omega, s, \mathbf{m})^T \cdot \mathbf{S}(\omega, \mathbf{m})^{-1} \cdot (\tilde{\mathbf{p}}(\omega, s, \mathbf{m}) - \tilde{\mathbf{d}}(\omega, s))^* \right\} \right] \quad (3)$$

where

$$\mathbf{f}_i^v(\omega, s, \mathbf{m}) = -\frac{\partial \mathbf{S}(\omega, \mathbf{m})}{\partial m_i} \cdot \tilde{\mathbf{p}}(\omega, s, \mathbf{m}) \quad (4)$$

In Equation (4),  $\mathbf{f}_i^v$  represents the virtual sources with respect to model parameters, and  $\mathbf{S}$  is the modeling operator in the frequency domain. The gradient direction can be preconditioned by the Hessian matrix and the suitable step-length is selected to reduce total errors iteratively. We used the pseudo-Hessian [21], which deals with the diagonal component of the Hessian matrix, for computational efficiency.

### 2.1.2. Reverse-Time Migration

The basic principle of RTM is that reflectors are located when the first arrival of downgoing wave coincides with the upgoing wave [22]. Conventional RTM can be calculated by zero-lag correlation between forward and backward wavefields [16]. The forward wavefield propagates in a forward time with the sources, which can be obtained by source wavelet estimation [23]. On the other hand, the backward wavefield propagates in reverse time, and the observed data at all receivers are used as the sources. Both wavefields are calculated by the numerical modelling based on the two-way wave equation.

RTM can also be represented as the first gradient of FWI [24]. The gradient-based conventional imaging condition can be defined as:

$$\mathbf{IC} = \sum_{\omega} \sum_s \text{Re} \left[ \frac{\mathbf{f}_i^v(\omega, s, \mathbf{m})^T \tilde{\mathbf{b}}(\omega, s)}{\mathbf{f}_i^v(\omega, s, \mathbf{m})^T \mathbf{f}_i^v(\omega, s, \mathbf{m})} \right], \quad (5)$$

where

$$\tilde{\mathbf{b}}(\omega, s) = \mathbf{S}^{-1} \cdot \tilde{\mathbf{d}}(\omega, s) \quad (6)$$

Here,  $\tilde{\mathbf{b}}$  is the backpropagated wavefield. The difference from the gradient of FWI is that the source of the backpropagated wavefield is not the residual between the observed and modeled data, but only the observed data. The autocorrelation of virtual sources produces normalizing terms that are used to compensate for the dependency on the receiver coverage and the source wavelet [25]. Because both the modelling wavefield and the virtual source are dependent on the model parameters, it is important to build a reasonable velocity model in the FWI process.

### 2.1.3. Four-Dimensional Seismic Monitoring

The concept of 4D seismic monitoring is to analyze the difference between two or more 3D seismic data recorded at different times. In the CCS project, the baseline survey is implemented before CO<sub>2</sub> injection, and additional surveys are followed during injection or after injection for detecting leakage [26]. An important assumption in monitoring is to maintain a stable exploration environment, including the configuration of sources and receivers to be as similar as possible. The difference in seismic data can be interpreted as a change mainly due to CO<sub>2</sub> when the design of the surveys at each time interval is similar. The RTM images obtained by each survey, according to the scenario of CO<sub>2</sub> injection can

be used as input data for monitoring, and the more CO<sub>2</sub> is injected, the more obvious the difference from the baseline survey becomes due to the distinct physical properties of CO<sub>2</sub>.

## 2.2. Improved Seismic Imaging with DAF

### 2.2.1. Principle of DAF

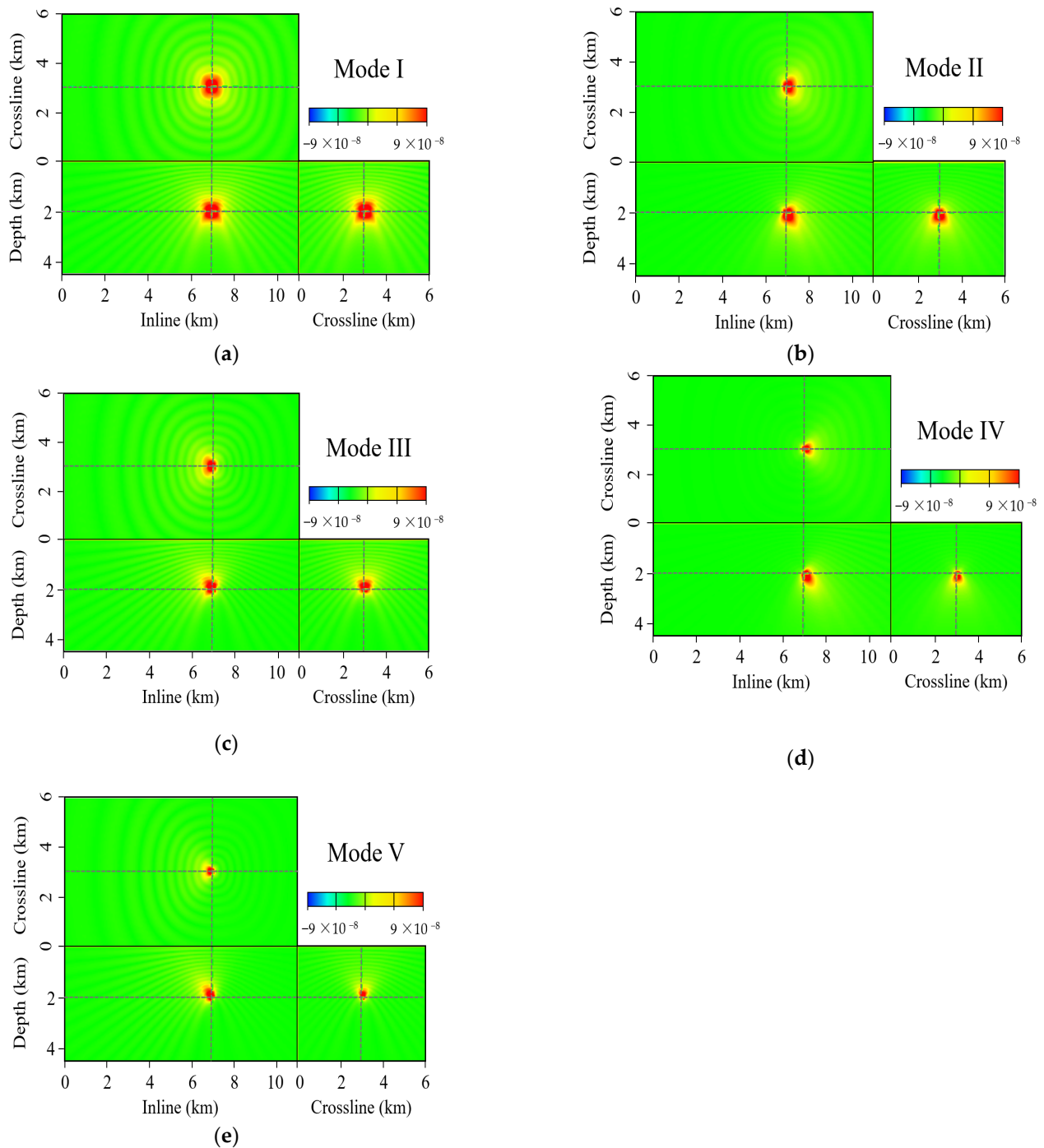
Diffracted seismic waves are generated by an anomaly including the boundary between layers, faults and fractures, and the results of FWI and RTM depend on the diffraction angle between incidence and diffracted rays [27]. DAF has been proposed to distinguish between the high- and low-wavenumber components of a gradient [19]. This technique works with a new virtual source by artificially adding terms from the virtual sources of the elastic wave equation. The modified virtual source with respect to the P-wave velocity ( $\mathbf{f}_i^{v,DAF}$ ) is calculated as follows [19]:

$$\begin{aligned} \mathbf{f}_i^{v,DAF}(\omega, s, \mathbf{m}) = & a_1 \nabla \left[ v_p (\nabla \cdot \tilde{\mathbf{u}}(\omega, s, \mathbf{m})) \right] + a_2 \left[ \frac{1}{v_p} \tilde{\mathbf{a}}(\omega, s, \mathbf{m}) \right] \\ & + a_1 \left( \frac{1-2\sigma}{1-2\sigma} \right) v_p \left[ \frac{\partial \tilde{\mathbf{s}}_A(\omega, s, \mathbf{m})}{\partial v_p} \right], \end{aligned} \quad (7)$$

where

$$\frac{\partial \tilde{\mathbf{s}}_A}{\partial v_p} = -2\nabla(\nabla \cdot \tilde{\mathbf{u}}) + \nabla \cdot [\nabla \cdot \tilde{\mathbf{u}} + (\nabla \cdot \tilde{\mathbf{u}})^T] - [\nabla \times (\nabla \times \tilde{\mathbf{u}})]. \quad (8)$$

$a_1$  and  $a_2$  are the weighting coefficients, and  $\sigma$  is Poisson's ratio.  $\mathbf{u}$  and  $\mathbf{a}$  represent the particle displacement and acceleration which are converted from pressure wavefields, respectively. The last term in Equation (8) become zero in acoustic media. Oh et al. (2021) classified DAFs into five modes by controlling the weighting coefficients. Mode I ( $a_1 = 2$ ,  $a_2 = 0$  and  $\sigma = 0.5$ ) derives the same gradient as the conventional FWI and RTM using all diffraction angles. Mode II ( $a_1 = 1$ ,  $a_2 = 1$  and  $\sigma = 0.5$ ) suppresses the diffraction energies at small and intermediate diffraction angles, while mode III ( $a_1 = 1$ ,  $a_2 = -1$  and  $\sigma = 0.5$ ) wipes out the diffracted energies at the large diffraction angle. Both mode II [28,29] and mode III [29,30] have the advantage of being able to use specific ranges of diffraction angles, but intermediate-wavenumber components are not completely eliminated. By adding a third term in Equation (7), mode IV ( $a_1 = 1$ ,  $a_2 = 1$  and  $\sigma = 0.25$ ) can choose large-angle components more clearly than mode II, and mode V ( $a_1 = 1$ ,  $a_2 = -1$  and  $\sigma = 0.25$ ) can focus more on small-angle components than mode III. Figure 2 shows the partial derivative wavefields using different modes of DAFs. The diffracted waves from mode I propagate in all directions. With mode II and IV, the diffraction energies at small and intermediate diffraction angles are suppressed, but the influence from intermediate-wavenumber components is further reduced in mode IV. With mode III and V, the diffracted wavefields at large diffraction angle are diminished, but the intermediate wavenumber components are more clearly missing in mode V.



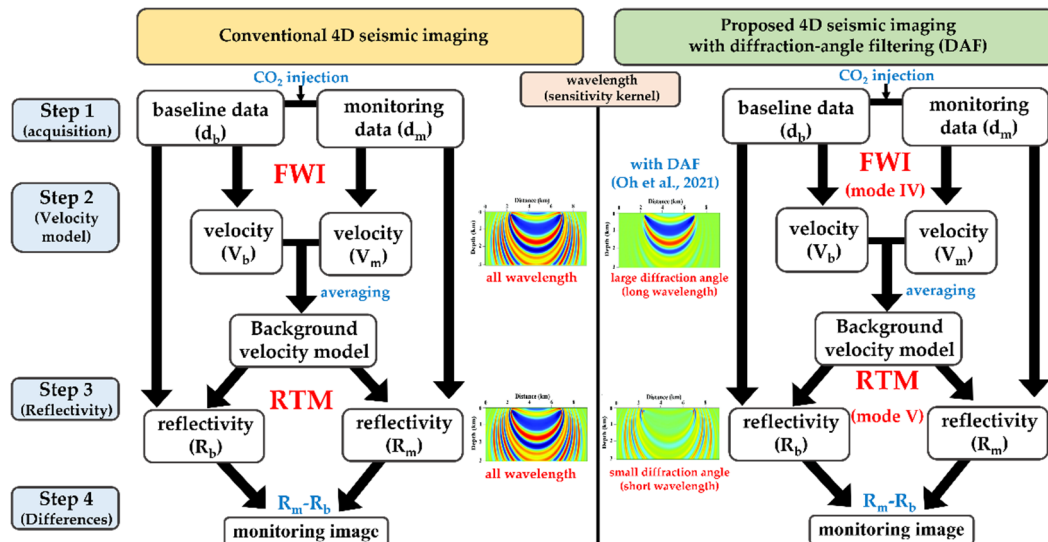
**Figure 2.** Snapshots of the partial derivative wavefields with different DAFs from mode I to V (a–e).

### 2.2.2. Workflow of Two-Step Monitoring with DAF

The goal of 4D seismic monitoring for CCS project is finding injected CO<sub>2</sub> plume by estimating the changes in two sets of seismic data acquired before and after injection, which are caused by velocity drops or reflectivity changes. Because FWI suffers from matching the amplitude information due to the non-repeatability of seismic surveys, we prefer to estimate reflectivity changes by CO<sub>2</sub> injection using migration techniques such as RTM after completing several preprocessing steps to mitigate repeatability issues. However, a background velocity model is still required for migration correctly to locate the CO<sub>2</sub> plume, and FWI can provide a high-resolution velocity model before RTM. The

4D seismic monitoring with FWI and RTM can be divided into the 4 main steps below (Figure 3):

- Step 1: 4D seismic data acquisition for baseline and monitoring data;
- Step 2: Velocity model building using FWI;
- Step 3: Reflectivity imaging using RTM with the background velocity model from Step 2;
- Step 4: Estimating CO<sub>2</sub> plume by subtracting two reflectivity images.



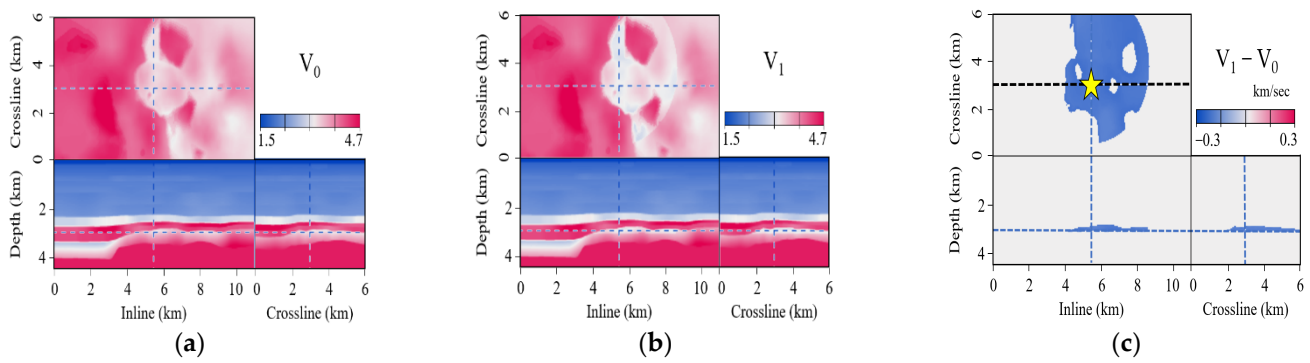
**Figure 3.** Workflow diagram of 4D seismic imaging with DAF [19] compared to conventional 4D seismic imaging.

As shown on the left side of Figure 3, FWI and RTM based on the sensitivity kernel utilized all information from the data, which contains all wavenumber components. However, we mainly require a long-wavelength background velocity model for RTM and short-wavelength reflectivity changes for high-resolution CO<sub>2</sub> plume. Thus, in the proposed two-step 4D seismic monitoring, we suggest applying DAF during FWI and RTM with different modes to separate long-wavelength and short-wavelength components for each procedure, respectively. Because DAF allows us to use the desired range of diffraction angles, mode IV of DAF can alleviate the local minima problem in FWI, and mode V helps to obtain the boundaries between layers more clearly.

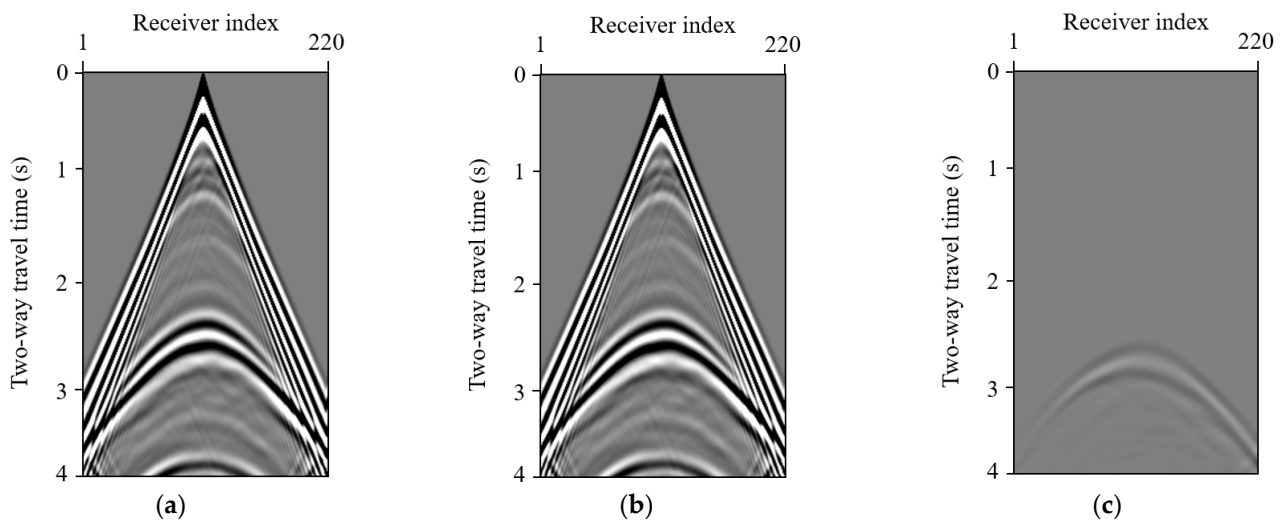
### 3. Results

#### 3.1. Site Description

We implement a synthetic test in the Volve oil field, which is in the Viking graben of the North Sea. A 3D ocean bottom cable (OBC) survey was conducted in 2010, and the seismic data and tomography models were released in 2018 by Equinor and its former Volve license partners. The Hugin Formation, which was deposited in the Jurassic period, is located at a depth of 3 km with caprocks [31]. We tested the 4D seismic monitoring approach in a scenario of injecting CO<sub>2</sub> in the Hugin Formation. Figure 4 shows the velocity model before and after the injection of the CO<sub>2</sub>. After the CO<sub>2</sub> injection, the velocity around the injection layer decreases by 6% compared to the baseline survey. The dimensions of the 3D velocity model were 12.3 × 6.8 × 4.5 km. Figure 5 shows the shot gathers before and after the injection of CO<sub>2</sub> and the subtraction between them. It is difficult to figure out the effect of injection on each shot gather, but the difference due to the CO<sub>2</sub> injection is clearly revealed in the subtraction between them.



**Figure 4.** P wave velocities (a) before and (b) after CO<sub>2</sub> injection in the Volve oil field. It was assumed that CO<sub>2</sub> was injected at depth of 3 km, and this can be confirmed in (c) the subtraction between the two velocity models. The yellow star and black dotted line in Figure 5 indicate the source and receiver location, respectively.



**Figure 5.** Shot gathers (a) before and (b) after injection, and (c) the subtraction between pre- and post-injection.

### 3.2. FWI Results

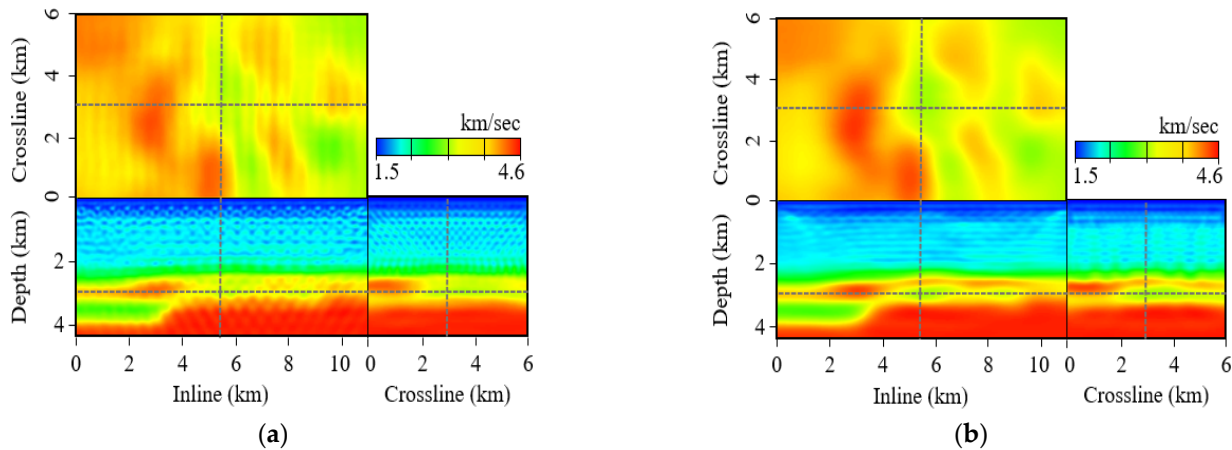
We performed 3D FWI in the Volve oil field using modes I and IV of DAF. Table 1 shows parameters used for FWI and RTM. The smoothed velocity models of Figure 4 are used as initial guesses and a Ricker wavelet is used for the source function. Due to the limitation of calculation capacity, the velocity was updated using a single frequency component (5 Hz) for 20 iterations. A total of 400 shots were uniformly distributed at the surface, and a total of 26,400 receivers were located at intervals of 50 m.

**Table 1.** Parameter information for FWI and RTM.

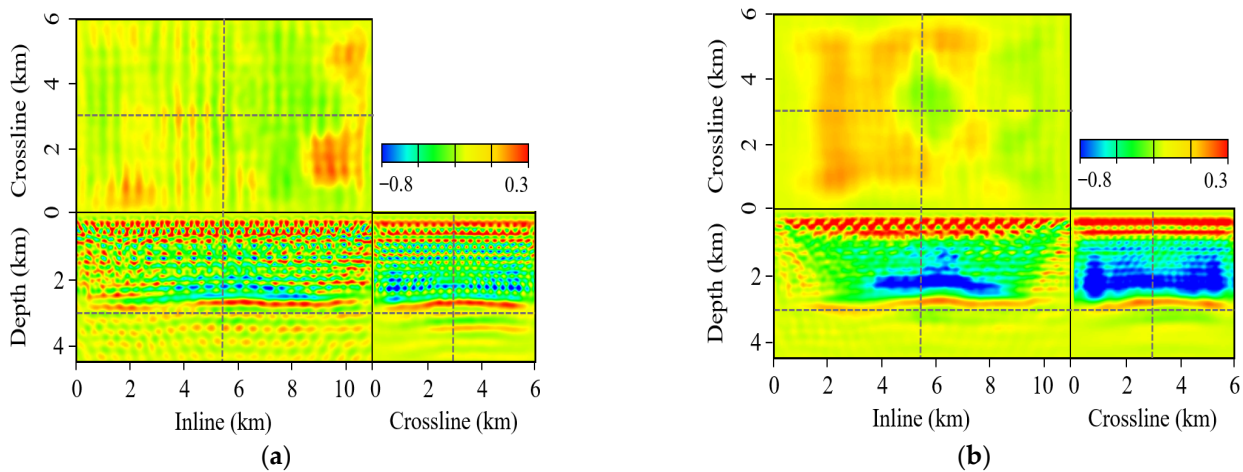
| Parameter            | FWI    | RTM |
|----------------------|--------|-----|
| Frequency (Hz)       | 5      | 3–8 |
| Number of iterations | 20     | 1   |
| Number of shots      | 400    |     |
| Number of recs       | 26,400 |     |
| Grid Sampling (m)    | 50     |     |

Figure 6 shows FWI results obtained by mode I and IV of DAF for the baseline survey. Because the diffracted wavefields of mode I include all wavenumbers, the target reservoir at 3 km was not properly located (Figure 6a). This means that a wrong gradient can

be constructed by the high wavenumber components before the update of long wavelength components. On the other hand, with mode IV, the target reservoir is well represented in the updated velocity due to the filtering of high wavenumber component (Figure 6b). Figure 7 shows the first gradient of FWI using mode I and IV. The gradient by mode I is skewed upward and shows weak signals around reservoir, while the low wavenumber components are well described, especially around the target reservoir because of designed filter in mode IV. Furthermore, noise generated by source and receiver configuration are mainly contained in the near-offset data, which hinders the inversion results in Figure 6a.



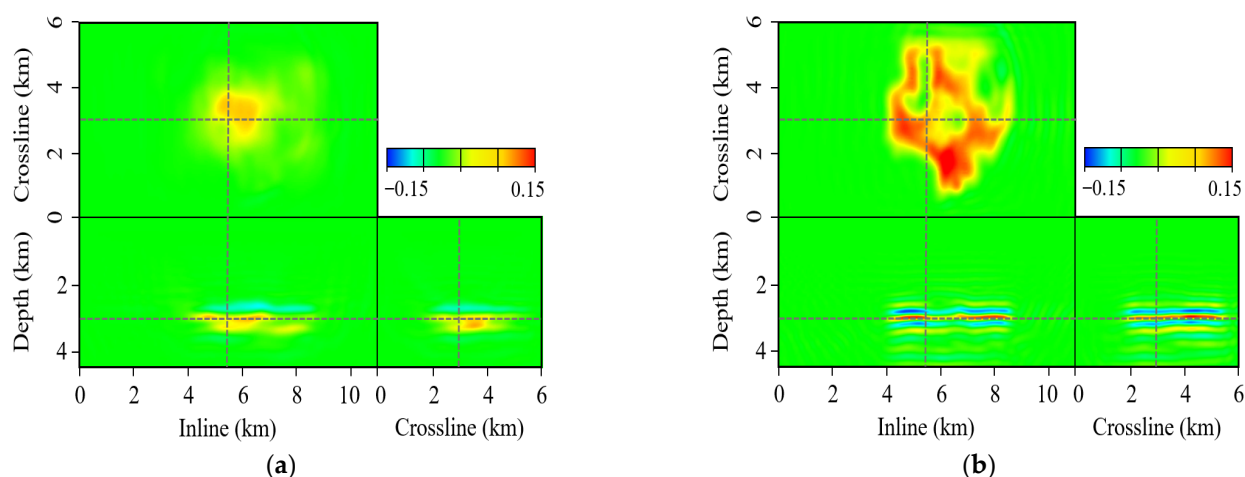
**Figure 6.** The inverted velocity model from FWI with (a) mode I (conventional) and (b) mode IV (large-angle) of DAF for baseline survey.



**Figure 7.** The first gradient direction of FWI using (a) mode I (conventional) and (b) mode IV (large-angle) of DAF.

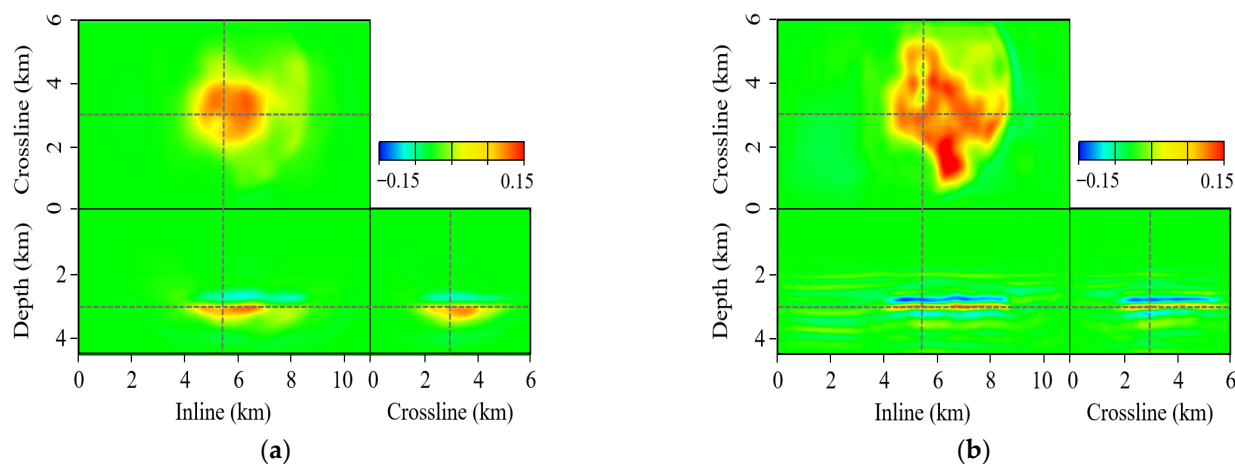
### 3.3. Monitoring Results by RTM Images

We performed RTM before and after the injection of CO<sub>2</sub>, respectively, and monitored the behavior of CO<sub>2</sub> through the subtraction between two images. The parameters used for RTM are presented in Table 1. Figure 8 shows the monitoring results using mode I and mode V of DAF with the conventional FWI result in Figure 6a. Because mode V of DAF filters out large scattering angles, the high wavenumber component mainly contributes to the correlation with forward wavefields. Thus, the subtraction with mode V represent sharper reflectors than conventional RTM images. However, the location of the CO<sub>2</sub> plume in both results is higher than that of the true model because the inaccurate velocity obtained from conventional FWI in Figure 6a was used.



**Figure 8.** Monitoring results by subtracting two RTM images from the baseline and post-injection survey with (a) mode I (conventional) and (b) mode V (small-angle) of DAF. The inverted velocity model (Figure 6a) using conventional FWI is used.

Figure 9 shows the monitoring results using mode I and mode V of DAF with the velocity model after FWI using mode IV of DAF in Figure 6b. Both monitoring approaches positioned the CO<sub>2</sub> plume more accurately than the conventional results in Figure 8, which indicate the importance of accuracy in the background velocity in RTM. However, conventional RTM uses all wavenumbers, so even with improved velocity, it cannot describe sharp reflectivities. The result using mode V of DAF showed a better extension of the reflectors and more accurate location of the CO<sub>2</sub> plume than that of conventional FWI. In addition, the amplitude difference in the inline-crossline plane of the depth of target is most the largest in Figure 9b, which means that the proposed method is the most suitable for 4D seismic monitoring.



**Figure 9.** Monitoring results by subtracting two RTM images from the baseline and post-injection survey with (a) mode I (conventional) and (b) mode V (small angle) of DAF. The inverted velocity model (Figure 6b) using FWI with mode IV of DAF is used.

#### 4. Discussion

The Volve oil field, which we used for our numerical test, has strong anisotropy above the caprock [31]. We performed a synthetic test so that the changes due to anisotropy did not affect the monitoring results, but a gap between modelling and real data may occur when acoustic assumptions are made. The elastic wave equation is suitable for describing the wave propagation in anisotropic media. Unfortunately, simulating in 3D using the elastic wave equation requires great computational cost, and several variables

increase its uncertainty. To compensate for these problems, an acoustic approximation has been suggested [32,33], and we will apply the acoustic approximation method to our scheme when treating real data in a future study.

The FWI and RTM of our test were calculated in the frequency domain. In general, the simulation in the frequency domain is efficient because the inverse matrix of the modelling operator can be applied to the sources in parallel, and we can analyze each frequency component depending on the purpose of survey. However, in the large exploration area, the huge size of the modelling operator matrix requires significant computational capacity, in which case performing the simulation in the time domain may be advantageous. Although we used a single frequency for FWI to reduce the computational burden, the updated velocity can be improved when the multi frequency components are gathered to construct the gradient.

In real seismic monitoring for CCS, it is important to match the exploration environment between the baseline and post-injection surveys. Because the post-injection survey is carried out after a certain period, the configuration of sources and receivers may change, and the properties of the seismic signal, such as its amplitude, are different for each survey. Therefore, in the post-injection survey, geometric correction is required for both sources and receivers to improve the low repeatability [34], and the processing technique that filters nearby disturbances from signals should be applied. From this point of view, our two-step monitoring strategy with DAF can contribute to obtaining robust monitoring results because specific wavenumber components are selected for velocity estimation with FWI and CO<sub>2</sub> detection with RTM, respectively.

## 5. Conclusions

We proposed a two-step seismic monitoring approach for CCS projects using DAF. In the first step, a velocity model is constructed using FWI with mode IV of DAF. Mode IV of DAF helps to prevent the local minima problem because the gradient of FWI is mainly composed of low-wavenumber components. In the second step, the seismic image is obtained by RTM with mode V of DAF using the updated velocity model through FWI, and the boundaries between layers are well represented due to the high-wavenumber components. The behavior of CO<sub>2</sub> can be monitored from the difference in seismic images between surveys performed before and after injection. To verify our method, we implemented synthetic testing using the velocity model in the Volve oil field. The long-wavelength structures around the reservoir were well constructed by FWI, and the CO<sub>2</sub> plume was detected in the correct position in the monitoring results using RTM.

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